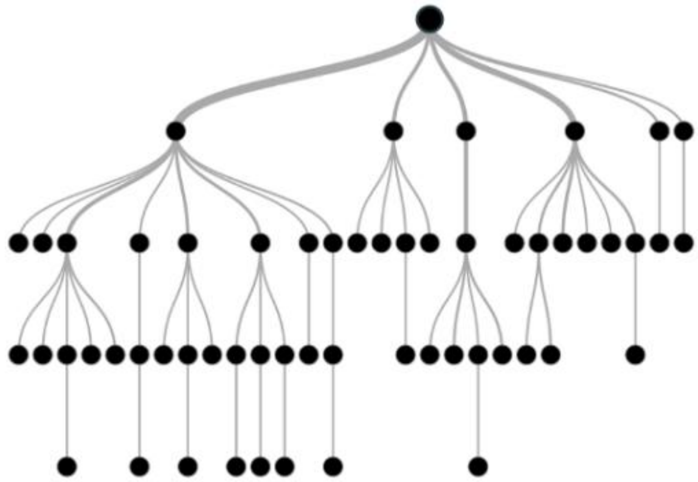
INTRODUCTION:

A decision tree is a supervised learning algorithm used for both classification and regression problems. It is represented as a tree structure where each internal node represents a test on an attribute, each branch represents the outcome of the test, and each leaf node represents a class label or a predicted value.

The goal of a decision tree is to split the dataset into subsets based on the value of an attribute, repeating this process until each subset contains only instances that belong to a single class or have similar values. It is a tool that has applications spanning several different areas. Decision trees can be used for classification as well as regression problems. The name itself suggests that it uses a flowchart like a tree structure to show the predictions that result from a series of feature-based splits. It starts with a root node and ends with a decision made by leaves.



Decision Tree Terminologies

- **Root Node**: This is the topmost node in a decision tree, representing the entire dataset. The root node is where the first split occurs based on the most significant attribute.
- **Leaf Node**: A leaf node is the terminal node that represents the final classification or output. It does not split further and provides the decision based on the previous splits.
- **Splitting**: This refers to the process of dividing a node into sub-nodes based on a certain condition or attribute. Each split corresponds to a decision made by the model.
- **Branch/Subtree**: A branch represents the subset of data that results from a split. It leads from a parent node to a child node or leaf.
- **Decision Node**: A node that represents a decision to split the data further based on a particular feature or attribute.
- **Pruning**: The process of removing parts of the tree that are not contributing to the model's accuracy in order to prevent overfitting.

Construction of Decision Tree

Building a decision tree involves several steps that ensure the final model is both accurate and interpretable. Here's a breakdown of the process:

1. Selecting the Best Attribute

The first step in building a decision tree is selecting the best attribute to split the dataset. This is done using criteria such as **information gain** or **Gini index**. The attribute that results in the highest gain (or lowest impurity) is chosen as the root node.

2. Recursive Splitting

Once the best attribute is selected, the data is split into subsets based on the value of that attribute. This process is repeated recursively for each subset, creating branches and sub-branches until the data is perfectly split or meets a stopping criterion (e.g., maximum depth or minimum number of instances per leaf).

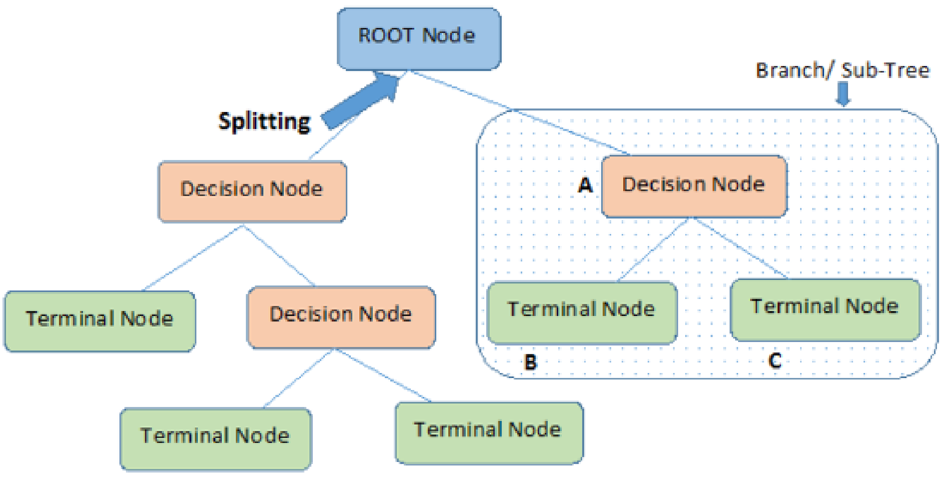
3. Stopping Criteria

The decision tree continues splitting the data until one of the following conditions is met:

- All instances in a node belong to the same class.
- The maximum depth of the tree is reached.
- The number of instances in a node falls below a specified threshold.

4. Tree Pruning

After the decision tree is fully grown, it may be necessary to prune it to remove overfitting. Pruning helps simplify the model by removing branches that do not contribute much to the accuracy of the model on unseen data.



Attribute Selection Measures

Choosing the right attribute to split the data at each node is a critical step in building an accurate decision tree. The most common methods used for attribute selection are **information gain** and **Gini index**.

Information Gain and Gini Index are metrics used to split nodes in decision trees, determining the best feature to maximize data purity. **Information Gain** measures the reduction in entropy (uncertainty) after a split, favoring attributes with higher gain. **Gini Index** measures impurity (misclassification probability), preferring smaller values for binary splits.

Information Gain (Entropy-based):

- **Concept:** Calculates the change in entropy from the parent node to the children nodes.
- **Formula:**

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v).$$

- **Usage:** Commonly used in ID3 and C4.5 decision tree algorithms.
- **Characteristics:** Tends to prefer features with more distinct values.

Gini Index (Impurity-based):

- **Concept:** Measures the probability of a random element being incorrectly classified if labeled according to the distribution in the subset.
- **Formula:**

$$\text{Gini} = 1 - \sum_{i=1}^{n} (P_i)^2, \text{ where } P_i \text{ is the probability of class } i.$$

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where P_i is the probability of class

- **Usage:** Used by the CART (Classification and Regression Trees) algorithm.
- **Characteristics:** Computationally faster as it avoids logarithmic calculations, creating binary splits.

Comparison of Information Gain and Gini Index

Feature	**Information Gain**	**Gini Index**
Criterion	Entropy (Uncertainty)	Impurity (Misclassification)
Algorithm	ID3, C4.5	CART
Calculation	Higher complexity (logarithm)	Lower complexity (no log)
Split Type	Multi-way or Binary	Binary
Formula	$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in Values(A)} \frac{\lvert S_v \rvert}{\lvert S \rvert} \text{Entropy}(S_v).$	$\text{Gini} = 1 - \sum_{i=1}^{n} (P_i)^2$, where P_i is the probability of class i.